REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR MOCK EXAMINATIONS- 2025

BIOLOGY 2C

MARKING GUIDE

(a) Procedures for preparation of solution S1

- The outer layer of onion bulb was removed
- Onion bulb is sliced or cut into small pieces (0.5marks@4= 2marks)
- The pieces were put or mixed into mortar and grinded by using pestle
- Little amount of water was added to the mixture and then filtered to get solution S1.

(b) FOOD TEST REPORT

FOOD TESTED	PROCEDURES	OBSERVATION	INFERENCE
Starch **(0.5mark)**	To 2cm3 of solution S1 in a test tube, 2 drops of iodine solution were added then shaken well (1mark)	The mixture retained the colour of iodine solution (1mark)	Starch was absent **(0.5mark)**
Protein **(0.5mark)**	To 2cm3 of solution S1 in a test tube, 2cm3 of sodium hydroxide solution was added, followed by 2 drops of 1% copper II sulphate solution, then shaken after each drop. (1mark)	The blue colour of copper II sulphate solution was retained (1mark)	Protein was absent **(0.5mark)**
Lipids **(0.5mark)**	To 2cm3 of solution S1 in a test tube, 3 drops of Sudan III solution were added then shaken vigorously and left to settle for five minutes (1mark)	Solution retained the colour of reagent (1mark)	Lipid was absent **(0.5mark)**
Reducing sugar **(0.5mark)**	To 2cm3 of solution S1 in a test tube, 2cm3 of Benedict's	The colour changed from blue, green, yellow, orange finally	Reducing sugar was present **(0.5mark)**

	solution was added then boiled (1mark)	to brick red observed(1mark)	
Non reducing sugar **(0.5mark)**	To 2cm3 of solution S1 in a test tube, 1cm3 of diluted Hydrochloric acid was added then shaken and boiled for two minutes then cooled, followed by addition of 2cm3 of Sodium hydroxide and 2cm3 (equal amount) of Benedict's solution then boiled (1mark)	The series of colour change observed from blue, green, yellow, orange finally to brick red (1mark)	Non reducing sugar was present **(0.5mark)**

(Total = 15marks)

(c) Food sample S1 has - Reducing sugar

- Non reducing sugar **(1@2 = 2marks**)

(d) Properties of reducing sugar and non-reducing sugars

- They are soluble in water
- They are insoluble in non-polar solvents **(Any 2@ 1= 2marks)**
- Have sweet taste or forming sweet tasting solutions.

(e) **ileum (1mark)**

(f) **Natural sources of identified food substances**

Milk, Honey and sugar cane **(2marks)**

(g) **Importance of identified food substance**

They provide energy to the body (Sources of energy to the body)

Reducing sugars are the building blocks of complex sugars **(Any 2@0.5= 1mark)**

Used for formation of sugars derivatives

2. (i) **(0.5@ 6 = 3marks**)

Organism	Common names

M	Scorpion
N	Spider
O	Yeast
P	Fern plant
Q	Ascaris
R	Cactus plant

(ii) Classifying organisms to class level

(0.5@ 15 = 7.5marks)

Organisms	Kingdom	Phylum/division	Class
M	Animalia	Arthropoda	Arachnida
N	Animalia	Arthropoda	Arachnida
O	Fungi	Ascomycota	Ascomycetes
P	Plantae	Filicinophyta	Filicinae
Q	Animalia	Nematoda	Chromadorea

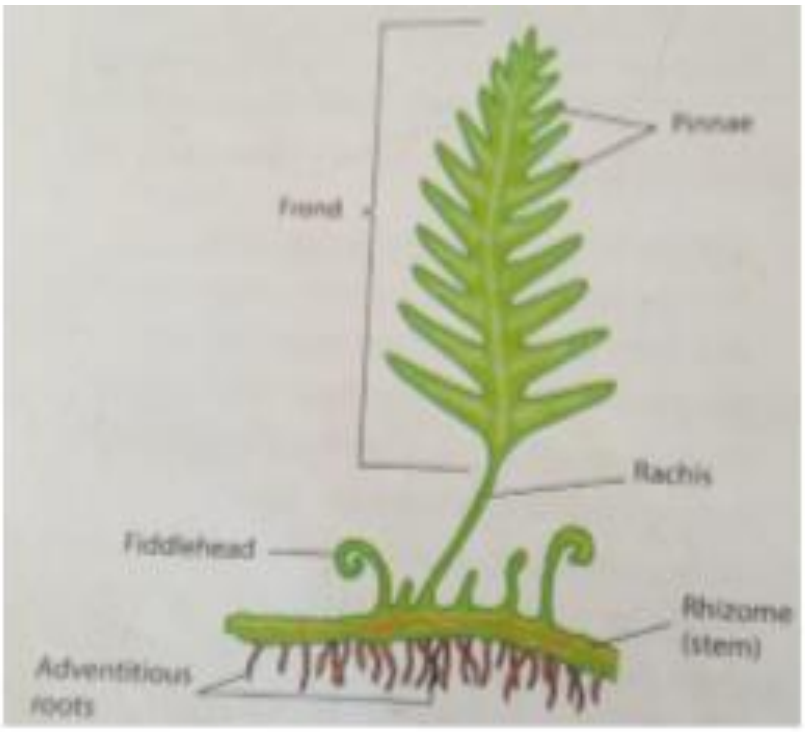
(iii) **Distinguishing features of phylum nematoda.**

- They have cylindrical bodies which taper at both ends.
- They are unsegmented round worms.
- They have longitudinal muscles; movement is restricted to dorsal ventral plane only.
- They have separate sex **(Any 4@0.5= 2marks)**

(iv) **General characteristics of Arachnida**

- They have two body parts i.e. Abdomen (opisthosoma) and cephalothorax (prosoma).
- They do not have antennae but they have a pair of pedipalp.
- They have four (4) pairs of legs (8 legs).
- Gaseous exchange occurs through the Book lungs
- Most have simple eyes **(Any 3@0.5= 1.5marks)**

(v) Diagram of a Fern plant



C=1mark

D= 2mark

L= 0.5@4= 2marks

(Total = 5marks)

(vi**) Adaptive features of cactus plants.**

- They have spiny leaves for reducing transpiration or water loss
- They have thick and fleshy stem for storage of water
- They have sunken stomata which minimize losing water by transpiration
- They have many root hairs for collecting and absorption of droplets of water
- They have waxy waterproofing coat for storage of water in the cactus
- Cactus have deeper tap root for anchoring the cactus and absorption of water

(Any 3@1= 3marks)

(vii) **Advantages of cactus in daily life**

- Cactus are used to prevent soil erosion
- Used to make drugs
- They provide oxygen in the atmosphere which is used by animals.
- Provide shelter and shadow to small animals
- They are used in scientific investigation and study

(Any 3@1= 3marks)

FORM FOUR MOCK EXAMINATIONS 2025

ACTUAL PRACTICAL BIOLOGY 2B

033/2B 3 HOURS ADVANCE INSTRUCTIONS

1.0 IMPORTANT

1.1 GREAT CARE MUST BE TAKEN NOT TO DIVULGE THESE INSTRUCTIONS TO BOTH CANDIDATES AND UNAUTHORIZED PERSONS, EITHER DIRECTLY OR INDIRECTLY

1.2 MAKE SURE THAT, THE CANDIDATES ARE PROVIDED WITH THE SPECIMENS, CHEMICALS AND APPARATI INDICATED IN THESE ADVANCE INSTRUCTIONS ONLY AND NOT OTHERWISE

1.3 PLEASE, DO NOT OPEN THE ENVELOPE CONTAINING THE QUESTION PAPERS,

3HOURS BEFORE THE TIME OF COMMENCEMENT OF THE EXAMINATION

A. PREPARATION OF SOLUTION

Onion bulb ……. S (01 per candidate

B. SPECIMENS

1. M …… Scorpion (01 per candidate)

2. N …… Spider (01 per candidate)

3. O ……. Yeast (1g per candidate)

4. P ……..…… Fern plant (01 shared by 4 candidates)

5. Q…… Ascaris (01 per candidate)

6. R……. Cactus plant (01 shared by 4 candidates)

C. TABLE REAGENT

1. Iodine solution

2. Sodium hydroxide solution

3. 1% Copper II sulphate solution (1%)

4. Sudan III solution

5. Benedict's solution

6. Dilute HCL solution

D. APPARATI

1. Knife …….…….…. (01 per candidate)

2. Sieve ……..……..……. (01 per candidate)

3. Beaker ……...…..…….……. (03 per candidate)

4. Test tubes ……..……..…… (05 per candidate)

5. Test tube rack ……..…… (01 per candidate)

6. Test tube holder ……..…… (01 per candidate)

7. Source of heat ……..…… (01 per candidate)

8. Test tube brush ……. (01 per candidate)

9. Mortar and pestle ……..…… (01 per candidate)

E. OTHER REQUIREMENTS

1. Distilled/rain water ……..……. (15m3 per candidate)

CHEK LIST FOR PRACTICAL 2025

BIOLOGY ACTUAL PRACTICALS

FOR BOTH SCHOOL AND PRIVATE CANDIDATES

1. SPECIMENS

Scorpion (01 per candidate)

Spider (01 per candidate)

Yeast (2g per candidate)

Fern plant (01 shared by 4 candidates)

Ascaris (01 per candidate)

Tomato fruits R (01 per candidate

Centipede (01 for 4 candidates)

Tick (01 per candidate)

Bread mould (1 per candidate)

Moss plant (01 per candidates)

Ascaris (01 per candidate)

Cactus plant (01 for 4 candidates

Onion bulbs (01 per candidate

Millipedes (01 per candidate)

Tilapia fish (01 per candidate)

Crab (01 per candidate)

2. APPARATUSES

Knife/scrapple (01 per candidate)

Sieve (01 per candidate)

Beaker 250mls (01 per candidate)

Test tubes (05 per candidate)

Test tube rack (01 per candidate)

Test tube holder (01 per candidate)

Source of heat (01 per candidate)

Test tube brush (01 per candidate)

Mortar and pestle (01 per candidate

Filter paper (01 per candidate)

Beaker 100mls

Measuring cylinder 100-200mls

Droppers

3. CHEMICALS AND REAGENTS

Iodine solution

Sodium hydroxide solution

1% Copper II sulphate solution (1%)

Sudan III solution

Benedict's solution

Dilute HCL solution

Distilled/rain water (15m3 per candidate

Glucose powder 200g per candidate

Sucrose powder 200g per candidate

Starch powder 200g per candidate

Egg albumen 100g per candidate

Cooking oils 200mls per candidate